Plancherel Theorem on the Real Line

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of plancherel theorem on the real line, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

HA-04 / Harmonic Analysis / Report v1

I. TARGET AND IMPORT CONTRACT

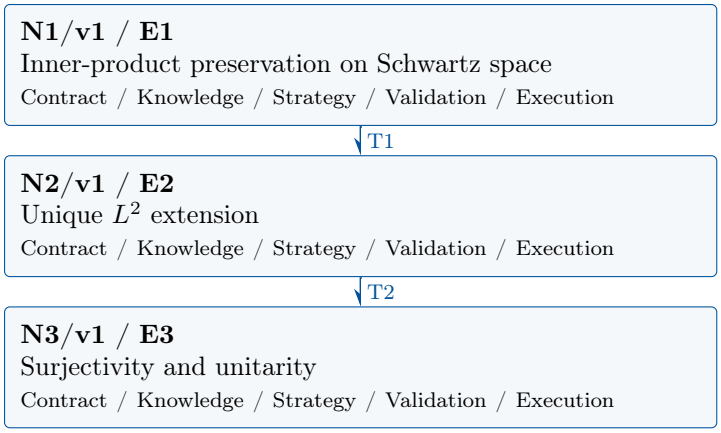
Theorem I.1. *With $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi ix\xi}\,dx$ on Schwartz functions, prove the transform extends uniquely to a unitary map on $L^2(\mathbb{R})$ preserving the L^2 norm.*

A. Foundations and permitted imports

A Schwartz function and all its derivatives decay faster than every power. Import Fourier inversion on $S(R)$ with convention $\widehat{f}(\xi) = \int f(x)e^{(-2\pi ix\xi)}dx$: $f(x) = \int \widehat{f}(\xi)e^{(2\pi ix\xi)}d\xi$. The transform and inverse preserve S, proved by differentiation and integration by parts. Schwartz functions are dense in $L^2(R)$; L^2 is complete. These are substantial supporting modules, explicitly imported rather than inferred from norm preservation.

Fubini is used only after the absolute-integrability bound $\|f\|_1\|\widehat{h}\|_1 < \infty$.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The Fourier transform preserves inner products and norms on S.

Receiving activity. N2: Unique L^2 extension

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Schwartz density and norm preservation; prior exports E1/v1

T2: N2/E2/v1 to N3. A unique linear isometric extension $U : L^2 \to L^2$ with closed range.

Receiving activity. N3: Surjectivity and unitarity

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Inversion on S and closed range; prior exports E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Inner-product preservation on Schwartz space

Object and role. Use inversion before extending the transform.

Inputs. $f, h \in S$; inversion and absolute integrability; foundational imports stated in §1

C — Contract

Own E1/v1: The Fourier transform preserves inner products and norms on S.

K — Knowledge

Retain $f, h \in S$; inversion and absolute integrability; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Use inversion before extending the transform.

V — Validation

Absolute integrability for Fubini is checked on this domain.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Inversion for h and conjugation give $\mathrm{conj}(h(x)) = \int \mathrm{conj}(\widehat{h}(\xi)) e^{(-2\pi i x \xi)} d\xi$.

N1.2 Since $\|f\|_1 \|\widehat{h}\,\|_1 < \infty$, Fubini applies to its product with $f(x)$.

N1.3 Thus $\int f(x)\,\mathrm{conj}(h(x))dx = \int [\int f(x) e^{(-2\pi i x \xi)} dx]\,\mathrm{conj}(\widehat{h}(\xi))d\xi = \int \widehat{f}(\xi)\,\mathrm{conj}(\widehat{h}(\xi))d\xi$.

N1.4 Taking $h = f$ proves $\|\widehat{f}\,\|_2 = \|f\|_2$ on S.

Export E1/v1. The Fourier transform preserves inner products and norms on S.

Acceptance gate. Absolute integrability for Fubini is checked on this domain. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Absolute integrability for Fubini is checked on this domain. Recheck dependent nodes N2, N3.

B. N2/v1: Unique L^2 extension

Object and role. Complete the isometry without using a pointwise integral for every L^2 function.

Inputs. Schwartz density and norm preservation; prior exports E1/v1

C — Contract

Own E2/v1: A unique linear isometric extension $U : L^2 \to L^2$ with closed range.

K — Knowledge

Retain Schwartz density and norm preservation; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Complete the isometry without using a pointwise integral for every L^2 function.

V — Validation

L^2 equivalence classes do not necessarily have an absolutely convergent Fourier integral.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For $f \in L^2$ choose $f_j \in S$ with $f_j \to f$ in L^2.

N2.2 The norm identity makes $\widehat{f_j}$ Cauchy, so it converges to an L^2 element Uf.

N2.3 A second approximating sequence gives the same limit by $\|\widehat{f_j} - \widehat{g_j}\|_2 = \|f_j - g_j\|_2$.

N2.4 This defines a linear isometry extending the transform, uniquely because S is dense.

N2.5 Its range is closed: a convergent sequence Uf_j makes f_j Cauchy, and its limit maps to the given range limit.

Export E2/v1. A unique linear isometric extension $U : L^2 \to L^2$ with closed range.

Acceptance gate. L^2 equivalence classes do not necessarily have an absolutely convergent Fourier integral. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: L^2 equivalence classes do not necessarily have an absolutely convergent Fourier integral. Recheck dependent nodes N3.

C. N3/v1: Surjectivity and unitarity

Object and role. Show the range contains a dense subset.

Inputs. Inversion on S and closed range; prior exports E2/v1

C — Contract

Own E3/v1: The transform extends uniquely to a unitary L^2 map.

K — Knowledge

Retain Inversion on S and closed range; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Show the range contains a dense subset.

V — Validation

Isometry alone does not imply onto; use dense range plus closedness.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Every $g \in S$ is the transform of its Schwartz inverse transform; hence S lies in the range of U.

N3.2 The range is closed and contains dense S, so it is all L^2.

N3.3 Inner-product preservation passes from Schwartz approximations to L^2 limits.

N3.4 Thus U is unitary with its inverse obtained by the same completion of the inverse transform.

Export E3/v1. The transform extends uniquely to a unitary L^2 map.

Acceptance gate. Isometry alone does not imply onto; use dense range plus closedness. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Isometry alone does not imply onto; use dense range plus closedness. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Inner-product preservation on Schwartz space

1. Inversion for h and conjugation give $\mathrm{conj}(h(x)) = \int \mathrm{conj}(\widehat{h}(\xi))e^{(-2\pi ix\xi)}d\xi$.
2. Since $\|f\|_1\|\widehat{h}\ \|_1 < \infty$, Fubini applies to its product with $f(x)$.
3. Thus $\int f(x)\,\mathrm{conj}(h(x))dx = \int[\int f(x)e^{(-2\pi ix\xi)}dx]\,\mathrm{conj}(\widehat{h}(\xi))d\xi = \int \widehat{f}(\xi)\,\mathrm{conj}(\widehat{h}(\xi))d\xi$.
4. Taking $h = f$ proves $\|\widehat{f}\ \|_2 = \|f\|_2$ on S.

B. §4.N2 — Unique L^2 extension

1. For $f \in L^2$ choose $f_j \in S$ with $f_j \to f$ in L^2.
2. The norm identity makes $\widehat{f_j}$ Cauchy, so it converges to an L^2 element Uf.
3. A second approximating sequence gives the same limit by $\|\widehat{f_j} - \widehat{g_j}\|_2 = \|f_j - g_j\|_2$.
4. This defines a linear isometry extending the transform, uniquely because S is dense.
5. Its range is closed: a convergent sequence Uf_j makes f_j Cauchy, and its limit maps to the given range limit.

C. §4.N3 — Surjectivity and unitarity

1. Every $g \in S$ is the transform of its Schwartz inverse transform; hence S lies in the range of U.
2. The range is closed and contains dense S, so it is all L^2.
3. Inner-product preservation passes from Schwartz approximations to L^2 limits.
4. Thus U is unitary with its inverse obtained by the same completion of the inverse transform.

□

V. WORKED ORCHESTRATION

For $f(x) = e^{(-\pi x^2)}$, the Gaussian integral calculation gives $\widehat{f} = f$ under this convention. Both squared norms equal $\int e^{(-2\pi x^2)}dx = 1/\sqrt{2}$. The inverse phase has the opposite sign; *retaining*2π avoids an erroneous normalization factor.

A. Boundary and hypothesis audit

The $function(1+|x|)^{(-3/4)}$ belongs to L^2 but notL [1]. Its Fourier transform is defined by completion, not an absolutely convergent integral at every frequency. Validation rejects a pointwise-integral export for all L^2 inputs and retains the equivalence-class contract.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

The $function(1+|x|)^{(-3/4)}$ belongs to L^2 but notL [1]. Its Fourier transform is defined by completion, not an absolutely convergent integral at every frequency. Validation rejects a pointwise-integral export for all L^2 inputs and retains the equivalence-class contract.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.